Calculus

❶ Integration by Parts

❷ Inverse Substitutions

❸ Improper Integrals

Example: evaluate $\int \tan x \, \mathrm{d}x$.

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$\int \tan x \, \mathrm{d}x = \int \frac{\sin x}{\cos x} \, \mathrm{d}x.$

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$\int \cos^4 ax \, dx = \int \left(\frac{1 + \cos 2ax}{2}\right)^2 dx$

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$\int \cos^4 ax \, dx = \int (\frac{1 + \cos 2ax}{2})^2 \, dx = \frac{1}{4} \int (1 + 2\cos 2ax + \cos^2 2ax) \, dx$

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$\int \cos^5 ax \, dx = \int (1 - \sin^2 ax)^2 \cos ax \, dx$. Let $u = \sin ax$, then $du = a \cos ax \, dx$, so $\int (1 - \sin^2 ax)^2 \cos ax \, dx = \frac{1}{a} \int (1 - u^2)^2 \, du = \frac{1}{a} \int (1 - 2u^2 + u^4) \, du = \frac{1}{a}(u - \frac{2}{3}u^3 + \frac{1}{5}u^5) + C = \frac{1}{a}(\sin ax - \frac{2}{3}\sin^3 ax + \frac{1}{5}\sin^5 ax) + C$

Example: evaluate $\int \cos^4 ax \, dx$.

$\int \cos^4 ax \, dx = \int (\frac{1+\cos 2ax}{2})^2 \, dx = \frac{1}{4} \int (1 + 2\cos 2ax + \cos^2 2ax) \, dx = \frac{1}{4} \int (1 + 2\cos 2ax + \frac{1+\cos 4ax}{2}) \, dx$

Example: evaluate $\int \tan x \, dx$.

$\int \tan x \, dx = \int \frac{\sin x}{\cos x} \, dx$. Let $u = \cos x$, then $du = -\sin x \, dx$. So $\int \frac{\sin x}{\cos x} \, dx = \int \frac{-1}{u} \, du = -\ln|u| + C = -\ln|\cos x| + C = \ln|\sec x| + C$.

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Example: evaluate $\int x^2 \sin x \, dx$.

Example: evaluate $\int x^2 \sin x \, \mathrm{d}x$.

Let $u = x^2$, then $\mathrm{d}u = 2x \, \mathrm{d}x$.

Example: evaluate $\int x^2 \sin x \, \mathrm{d}x$.

Let $u = x^2$, then $\mathrm{d}u = 2x \, \mathrm{d}x$. Let $\mathrm{d}v = \sin x \, \mathrm{d}x$, then $v = -\cos x$.

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To evaluate $\int x \cos x \, \mathrm{d}x$, let $u = x$, then $\mathrm{d}u = \mathrm{d}x$.

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Example: evaluate $\int x^2 \sin x \, dx$.

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Example: evaluate $\int e^{ax} \cos bx \, dx$ and $\int e^{ax} \sin bx \, dx$.

Example: evaluate $\int x^2 \sin x \, \mathrm{d}x$.

Let $u = x^2$, then $\mathrm{d}u = 2x \, \mathrm{d}x$. Let $\mathrm{d}v = \sin x \, \mathrm{d}x$, then $v = -\cos x$. So $\int x^2 \sin x \, \mathrm{d}x = x^2 \cdot (-\cos x) + 2 \int x \cos x \, \mathrm{d}x$

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To evaluate $\int e^{ax} \cos bx \, \mathrm{d}x$, Let $u = \cos bx$, then $\mathrm{d}u = -b \sin bx \, \mathrm{d}x$.

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Exercises

$$\int (x+3)e^{2x}\,\mathrm{d}x$$

$$\int (x+3)e^{2x}\,\mathrm{d}x = -\frac{(x+3)e^{2x}}{2} - \frac{e^{2x}}{4} + C$$

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$$\int (x^2-2x)e^{kx}\,\mathrm{d}x$$

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$$\int x(\ln x)^3\,\mathrm{d}x$$

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Inverse Substitutions

Improper Integrals

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